

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

WAVE MOTION

LECTURE-03

PHYSICS

WAJAHID SIR (Gabbar Sir)



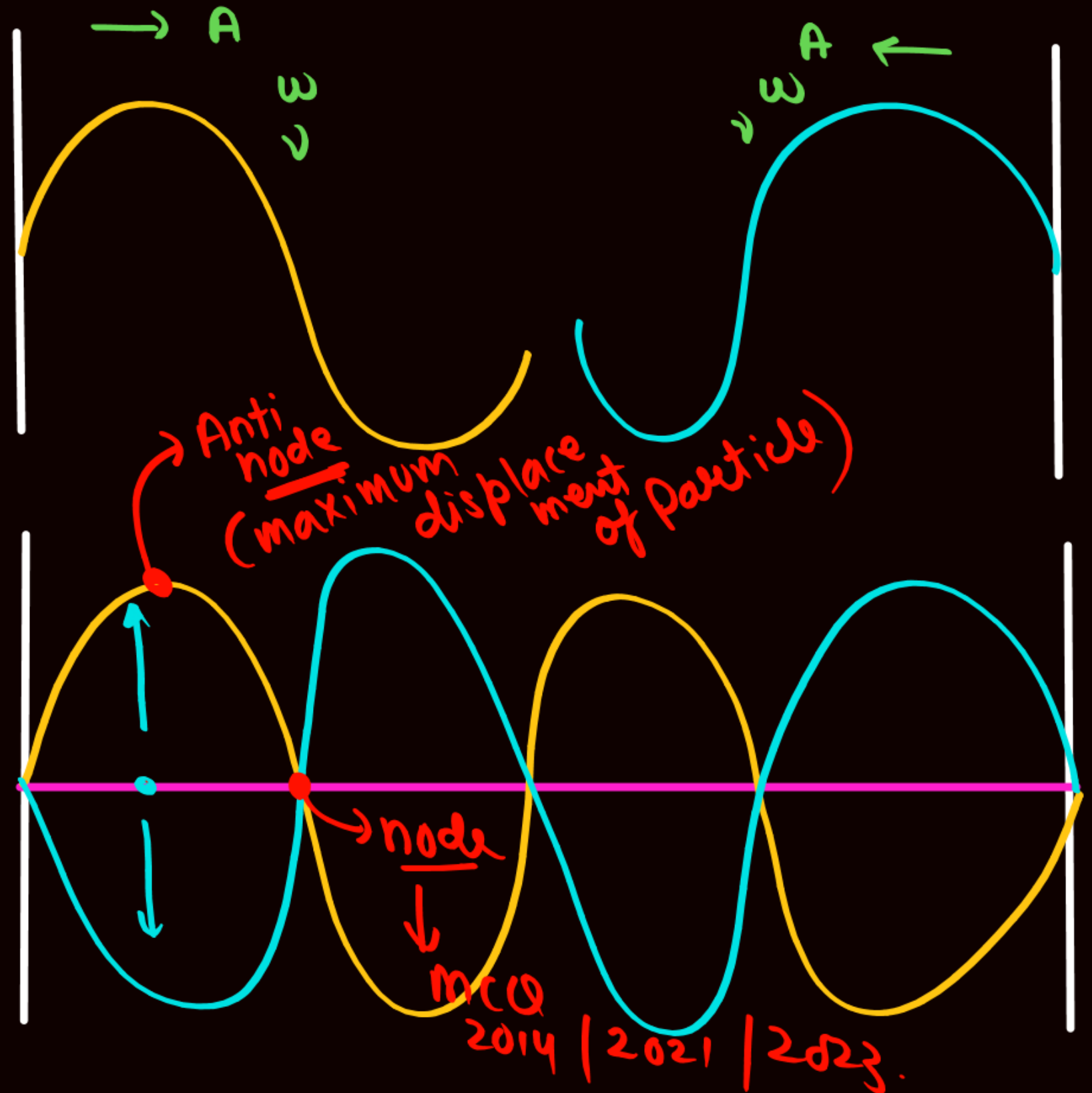
Topics To Be Covered

- 1 Revision dec- 1 /02
- 2 ✓ Standing Waves (5mark) ✓
 Part 2
 2026
 2025
 2024
 2023
- 3 ..
- 4 ..
- 5 ..

Standing wave

- when two waves of same Amplitude (A) and same Angular frequency (ω) Superimposed they form Standing wave

String
Organ pipe



Standing wave

(i) Standing wave in String (Bind at one end) $\rightarrow$ frequency $\textcircled{2}$

(ii) Standing wave in String (Bind at Two end) $\textcircled{3}$

(i) Standing wave in Close organ pipe

$$v_1 = \frac{1V}{4l}$$

$$v_2 = \frac{3V}{4l}$$

$$v_3 = \frac{5V}{4l}$$

(ii) Standing wave in Open organ pipe

$$v_1 = \frac{1V}{2l}$$

$$v_2 = \frac{2V}{2l}$$

$$v_3 = \frac{3V}{2l}$$

STANDING WAVE In a STRING Bind at one end.

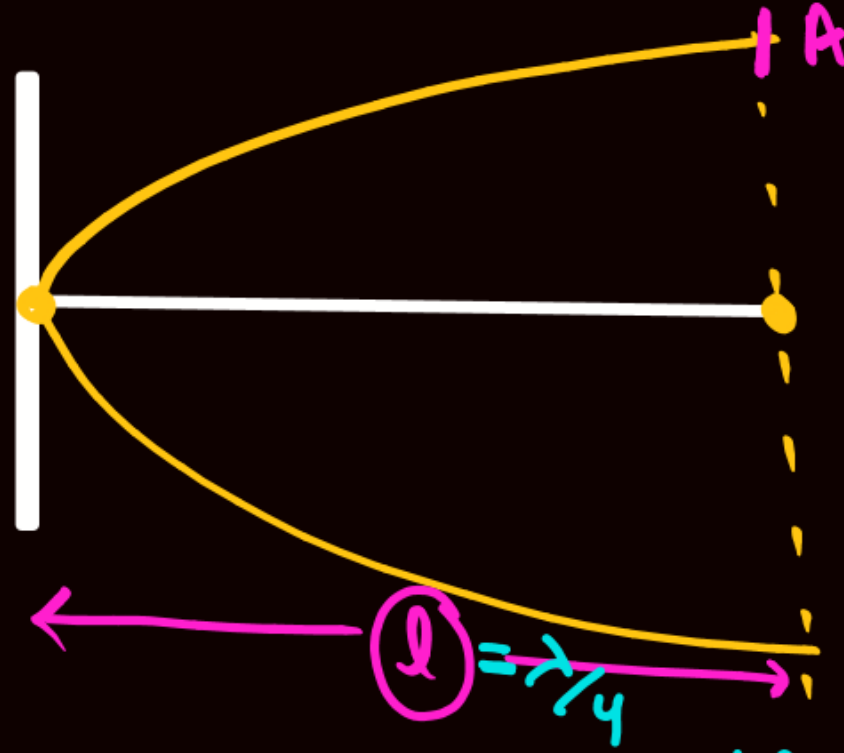
OR

STANDING WAVE In a closed organ pipe

③

(i) Ist Case (Ist mode of vibration)

$$l = \lambda/4$$



$$4l = \lambda$$

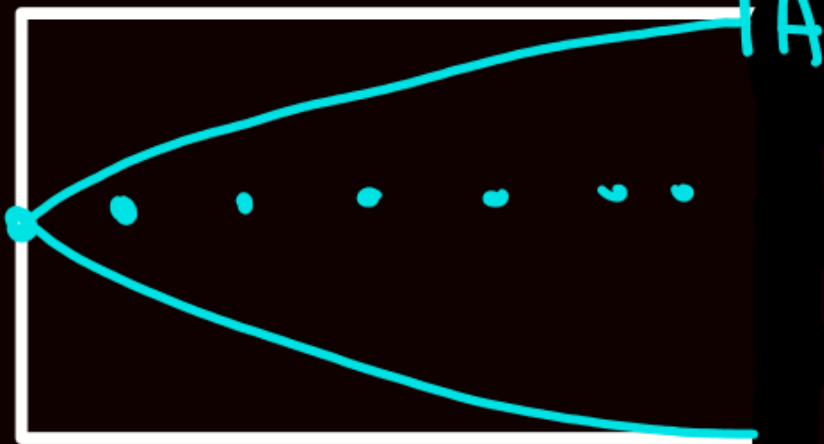
$$v = \lambda \nu$$

$$\frac{v}{\lambda} = \nu_1$$

$$\frac{v}{4l} = \nu_1$$

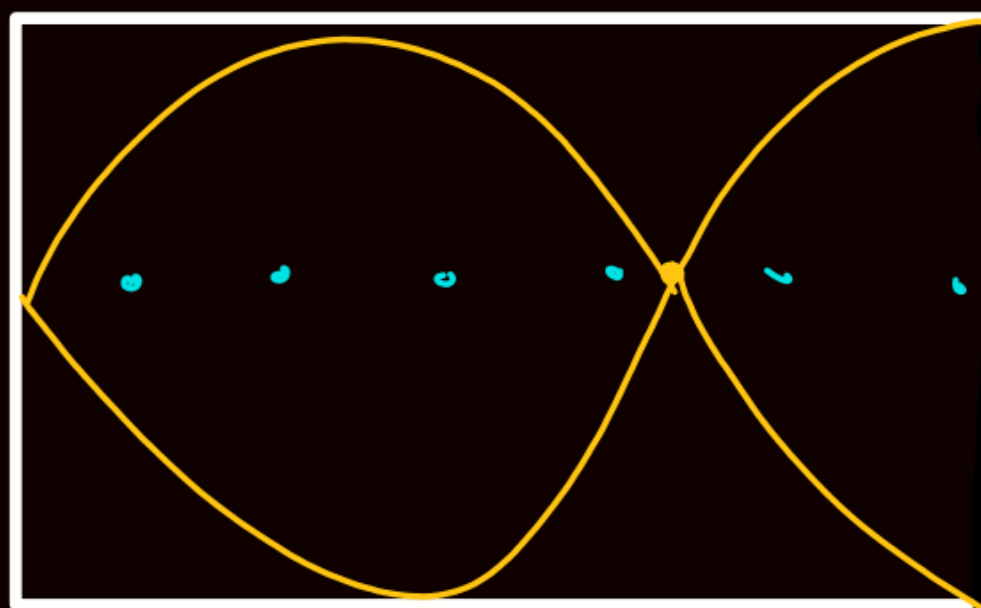
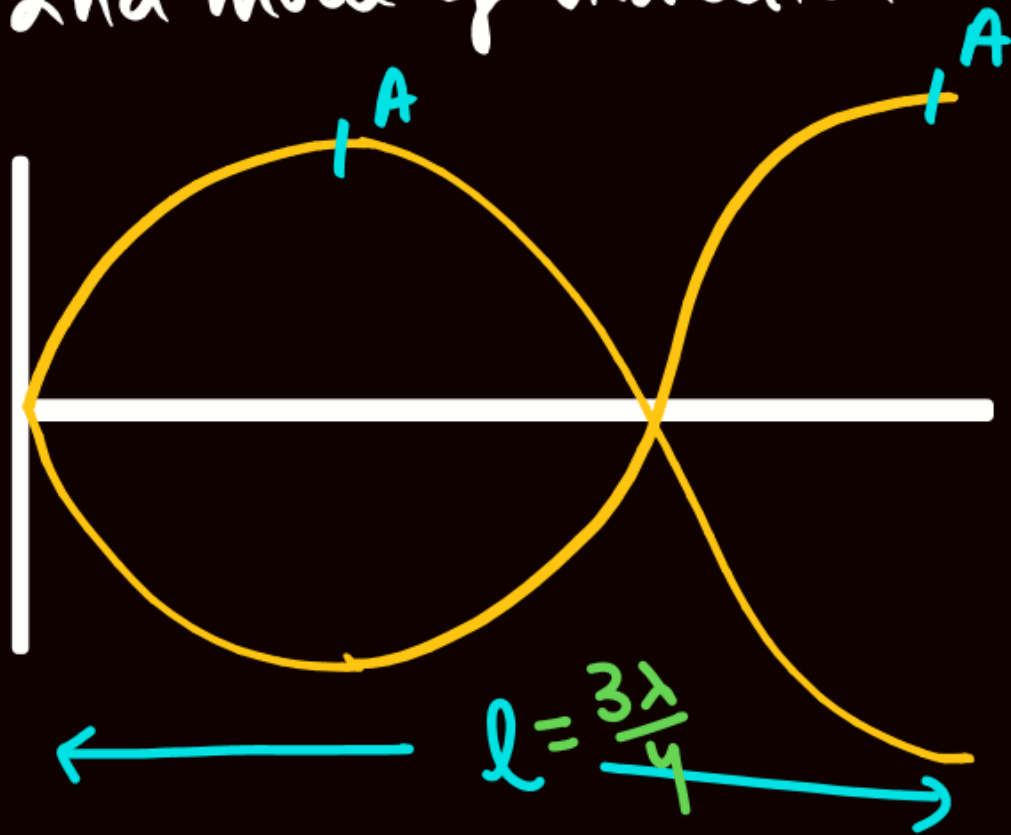
$$\nu_1 = \frac{v}{4l}$$

Ist frequency / Fundamental frequency.
Ist tone
Ist harmonic



(ii) 2nd case

2nd mode of vibration.



$$l = \frac{3\lambda}{4}$$

$$4l = 3\lambda$$

$$\frac{4l}{3} = \lambda$$

$$v = \lambda \nu_2$$

$$\frac{v}{\lambda} = \nu_2$$

$$\frac{v}{\frac{4l}{3}} = \nu_3$$

$$\frac{3v}{4l} = \nu_2 \quad \nu_1 \quad (4)$$

$$\nu_2 = 3 \left(\frac{v}{4l} \right)$$

$$\nu_2 = \frac{3v}{4l}$$

→ 2nd frequency

3rd tone

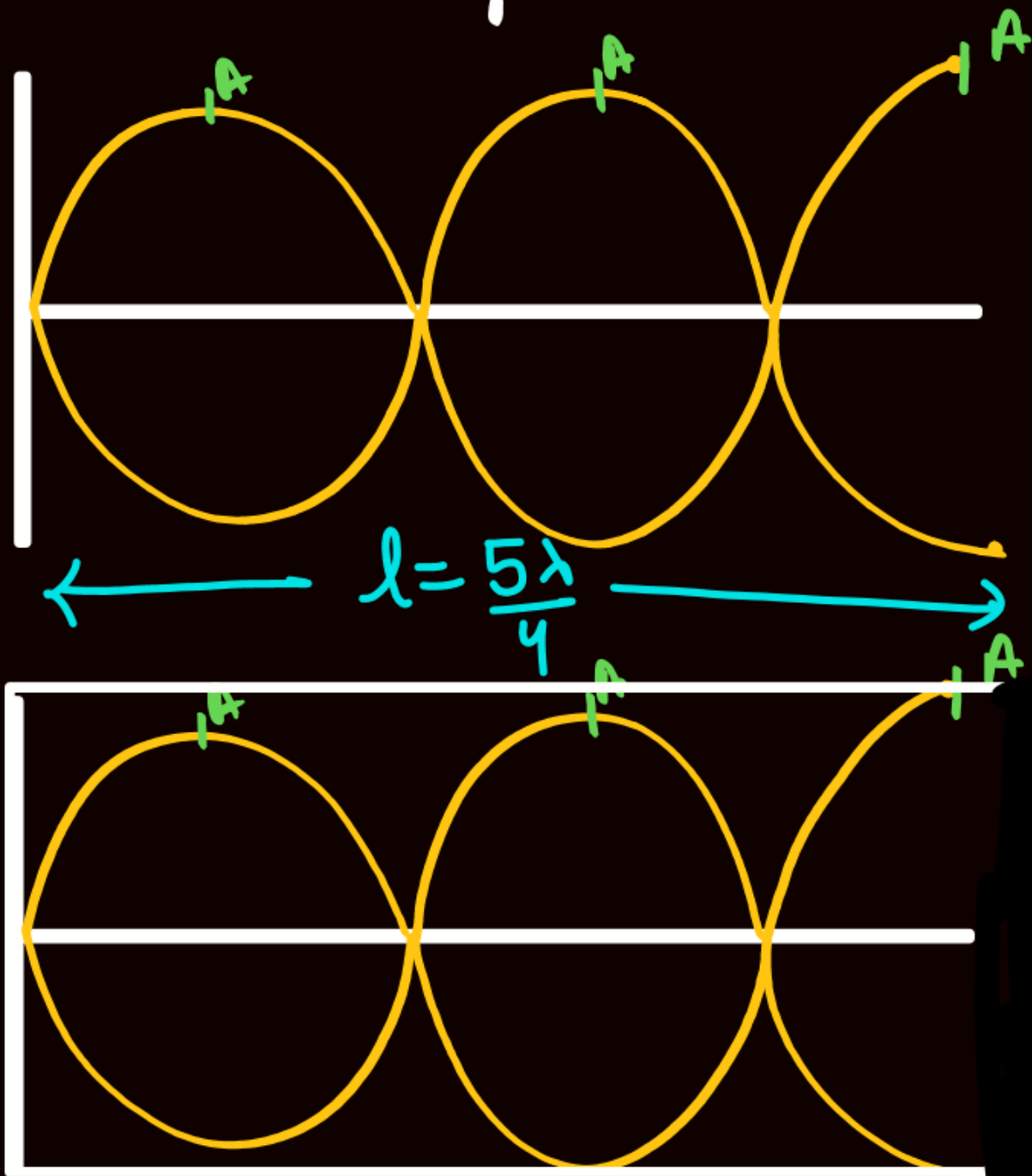
3rd harmonic

$$\nu_2 = 3 \left(\frac{v}{4l} \right)$$

$$\nu_2 = 3\nu_1$$

(iii) Case - III

3rd mode of vibration.



$$l = \frac{5\lambda}{4}$$

$$4l = 5\lambda$$

$$\frac{4l}{5} = \lambda$$

$$\bullet v = \lambda \nu_3$$

$$\frac{v}{\lambda} = \nu_3$$

$$\frac{v}{\frac{4l}{5}} = \nu_3$$

$$\boxed{\frac{5v}{4l} = \nu_3}$$

⑤

- 3rd frequency
- 5th tone
- 5th Harmonic

$$\nu_3 = 5 \left(\frac{v}{4l} \right) \nu_1$$

$$\nu_3 = 5\nu_1$$

Note

$$v_1 = \frac{v}{4l} = v_1$$

$$v_2 = 2 \left(\frac{v}{4l} \right) = 2v_1$$

$$v_3 = 3 \left(\frac{v}{4l} \right) = 3v_1$$

$$\frac{v_1}{v_2} = \frac{1}{2} \quad \frac{v_2}{v_3} = \frac{2}{3} \quad \frac{v_1}{v_3} = \frac{1}{3}$$

Even no. of harmonics
are missing out.

1:2:3

odd no.

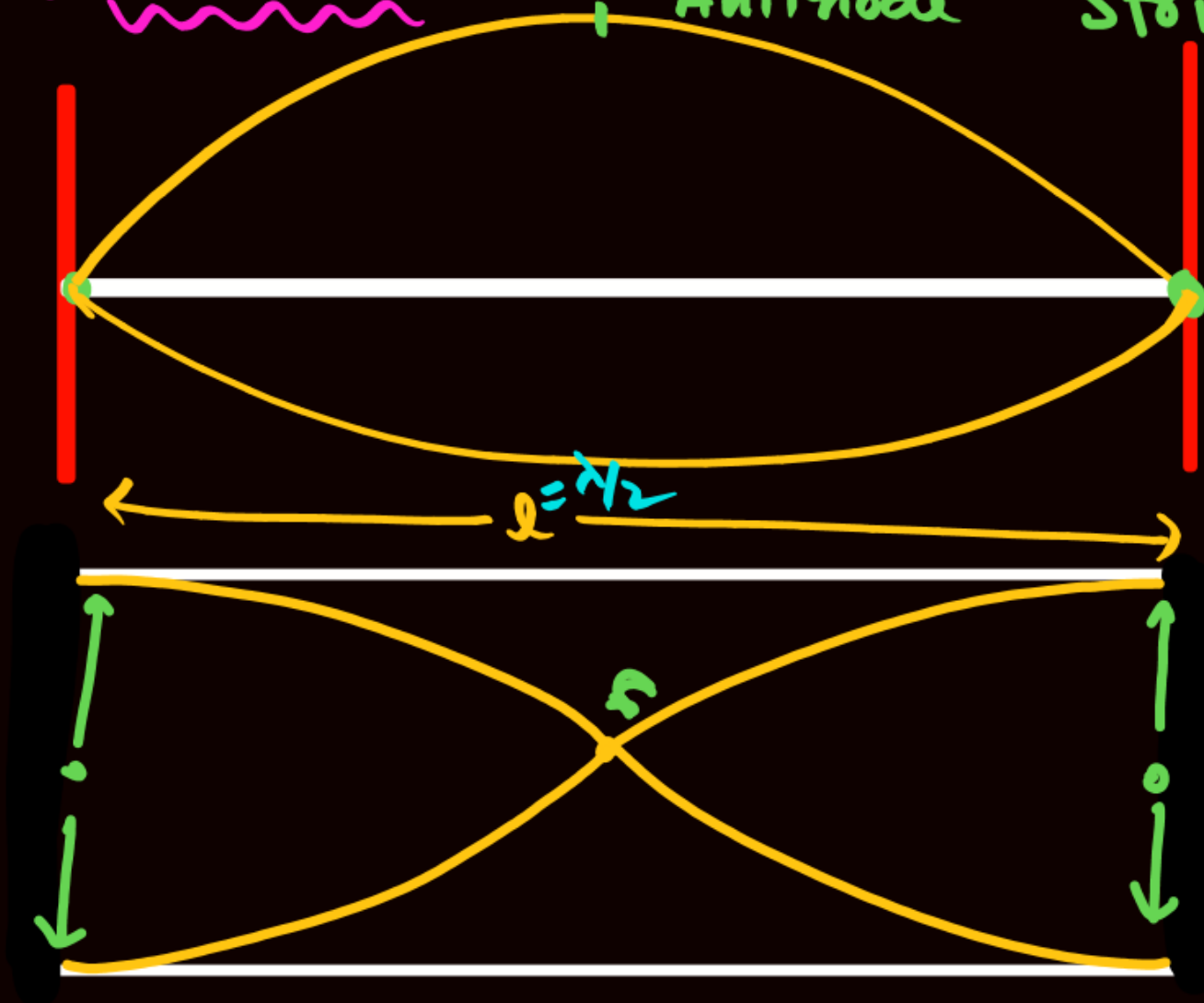
Ratio

STANDING WAVE IN STRING BIND AT TWO END

OR

STANDING WAVE IN OPEN ORGAN PIPE.

(1) Ist Case (Ist mode of vibration)



$$l = \frac{\lambda}{2}$$

$$2l = \lambda$$

$$\lambda = 2l$$

$$v = \lambda \nu$$

$$\frac{v}{\lambda} = \nu_1$$

$$\frac{v}{2l} = \nu_1$$

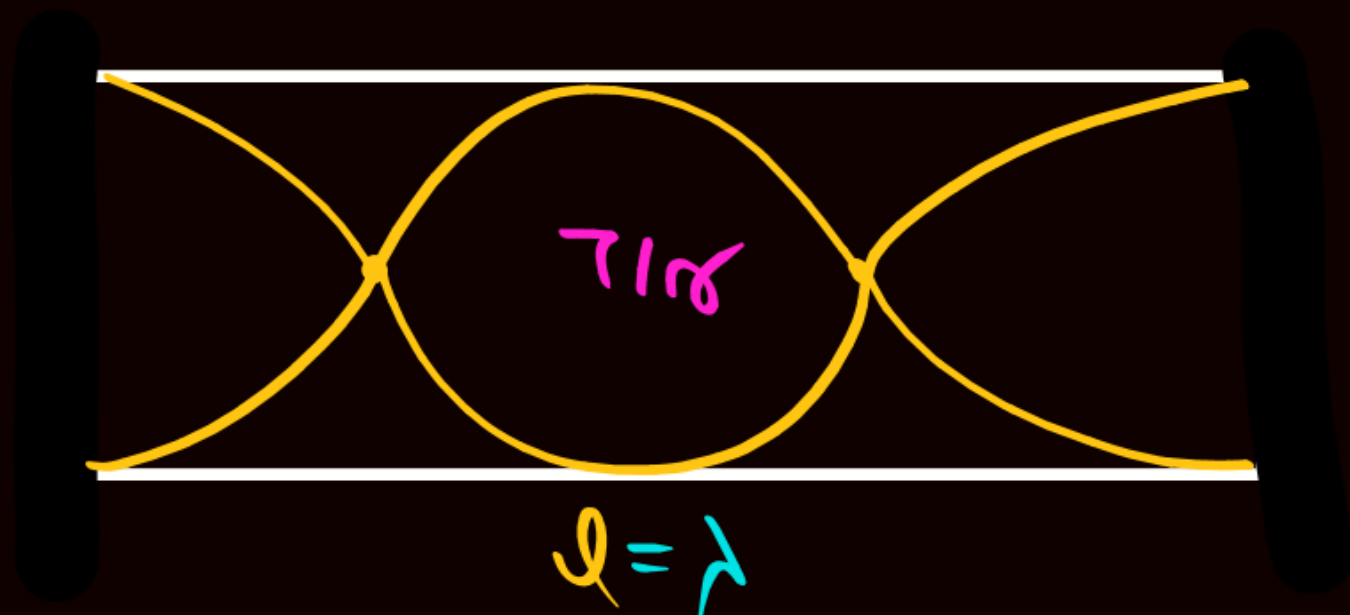
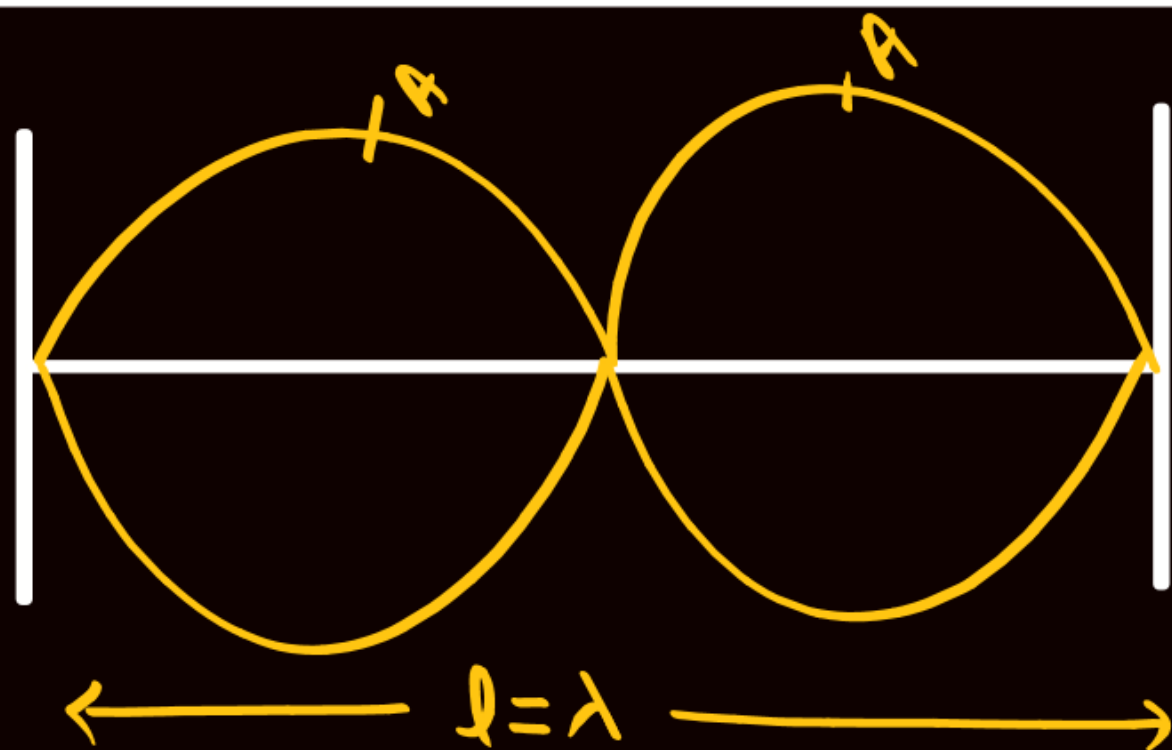
$$\nu_1 = \frac{v}{2l}$$

Ist frequency

Ist tone

Ist Harmonic

2nd case (2nd mode of vibration)



$$\lambda = l$$

$$v = \lambda \nu_2$$

$$\frac{v}{\lambda} = \nu_2$$

$$\frac{v}{l} = \nu_2$$

$$\nu_2 = \frac{v}{l}$$

$$\nu_2 = \frac{2v}{2l}$$

2nd frequency

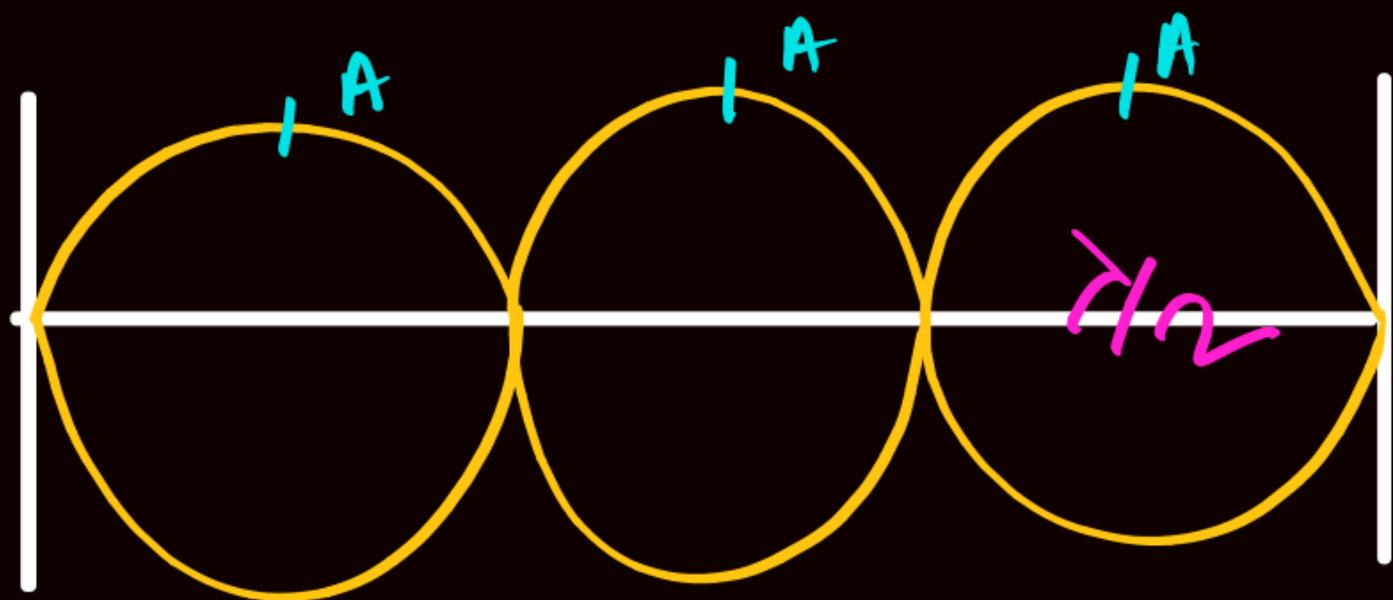
2nd tone

2nd harmonic

$$\nu_2 = 2 \left(\frac{v}{2l} \right) \nu_1$$

$$\nu_2 = 2\nu_1$$

3rd case (3rd mode of vibration)



$$l = \frac{3\lambda}{2}$$

$$l = \frac{3\lambda}{2}$$

$$2l = 3\lambda$$

$$\frac{2l}{3} = \lambda$$

$$v = \lambda \nu_3$$

$$\frac{v}{\lambda} = \nu_3$$

$$\frac{v}{\frac{2l}{3}} = \nu_3$$

$$\boxed{\frac{3v}{2l} = \nu_3}$$

$$\nu_3 = \frac{3v}{2l}$$

3rd frequency

3rd tone

3rd harmonic

$$\nu_3 = 3 \left(\frac{v}{2l} \right) \nu_1$$

$$\nu_3 = 3\nu_1 \checkmark$$

Note

$$v_1 = \frac{v}{2l} = v_1$$

$$v_2 = 2 \left(\frac{v}{2l} \right) = 2v_1$$

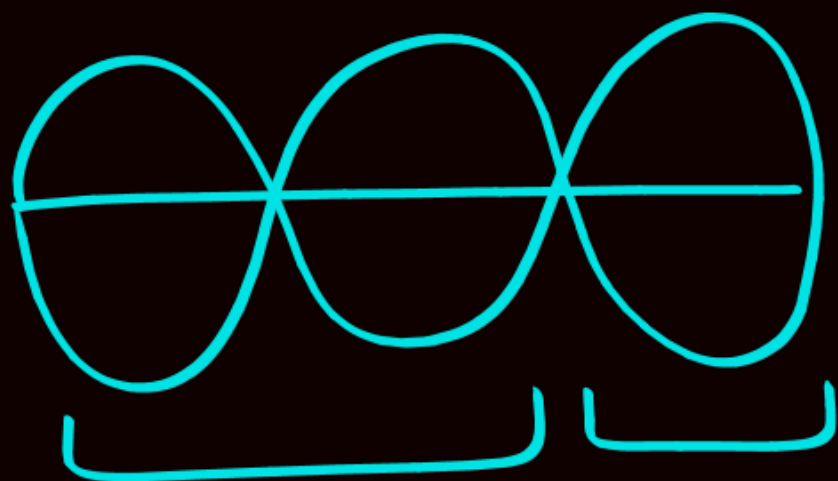
$$v_3 = 3 \frac{v}{2l} = 3v_1$$

$$\frac{v_1}{v_2} = \frac{1v_1}{2v_1} = \frac{1}{2}$$
$$\frac{v_2}{v_3} = \frac{2v_1}{3v_1} = \frac{2}{3}$$

1° 2° 3

No Harmonic

No tone is missing ✓

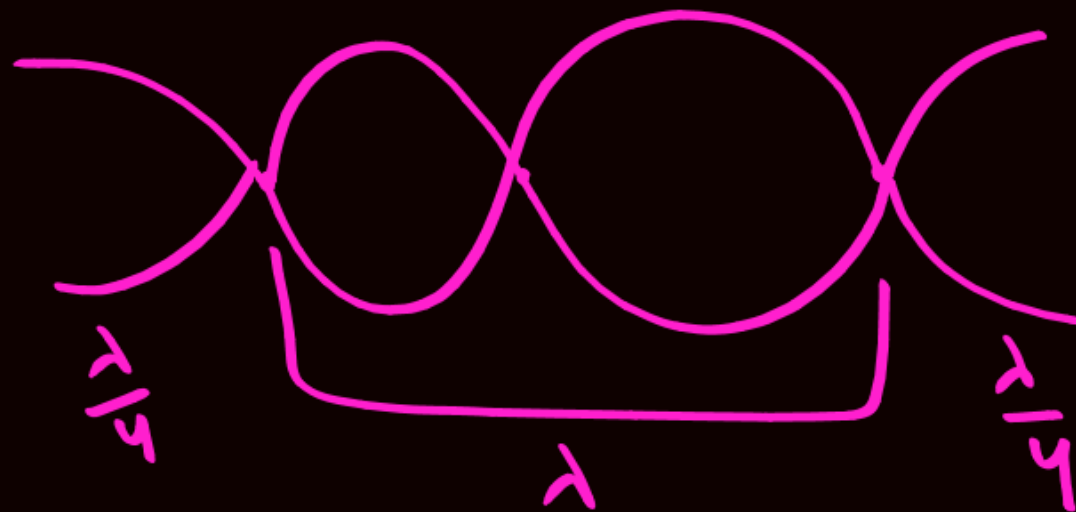


$$\lambda + \frac{\lambda}{2}$$

$$\lambda + \frac{\lambda}{2}$$

$$\frac{2\lambda + \lambda}{2}$$

$$l = \frac{3\lambda}{2}$$



$$\frac{\lambda}{4} + \frac{\lambda}{4} + \lambda$$

$$\frac{\lambda + \lambda}{4} + \lambda$$

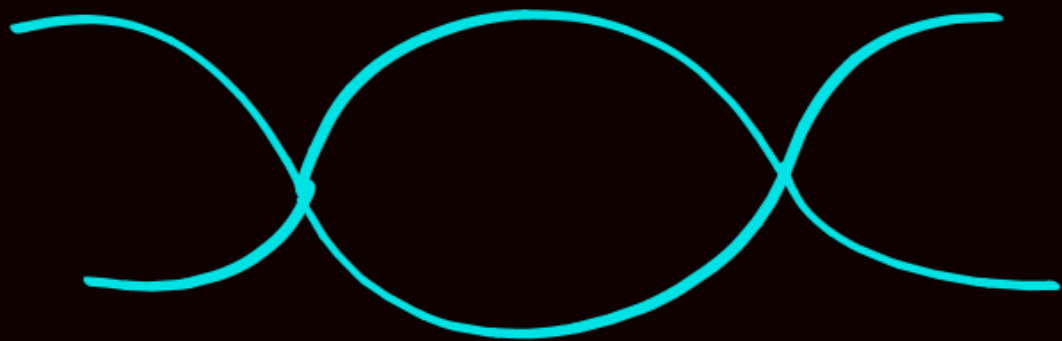
$$\frac{2\lambda}{4} + \lambda$$

$$\frac{\lambda}{2} + \lambda$$

$$\frac{\lambda}{2} + \lambda$$

$$\frac{\lambda + 2\lambda}{2}$$

$$l = \frac{3\lambda}{2}$$



$$\frac{\lambda}{4} + \frac{\lambda}{2} + \frac{\lambda}{4}$$

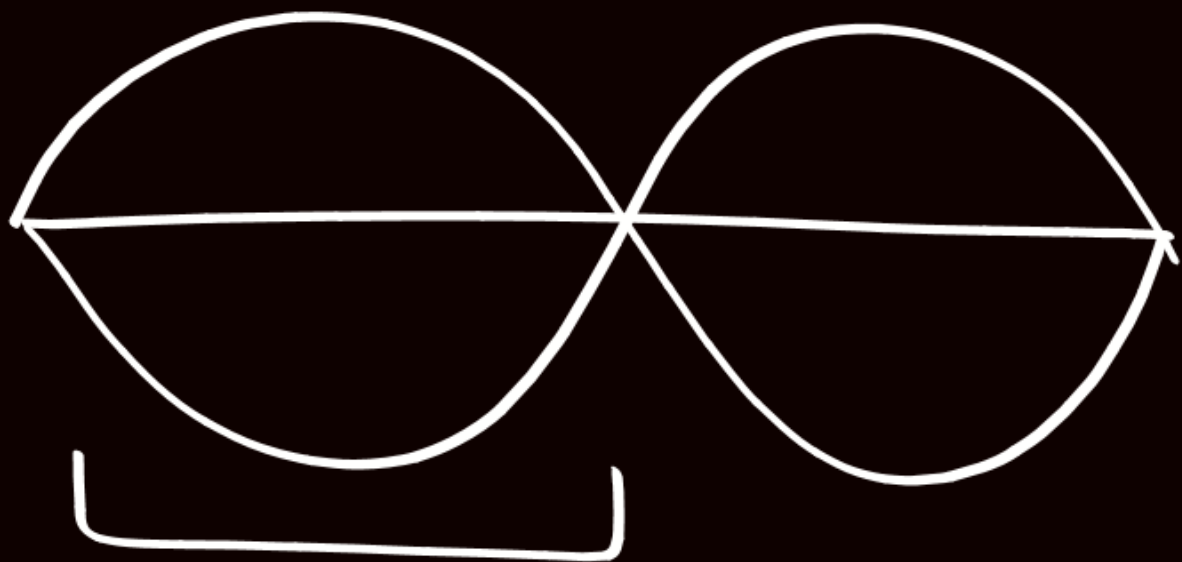
$$\frac{\lambda}{2} + \frac{\lambda}{4} + \frac{\lambda}{4}$$

$$\frac{\lambda}{2} + \frac{\lambda + \lambda}{4}$$

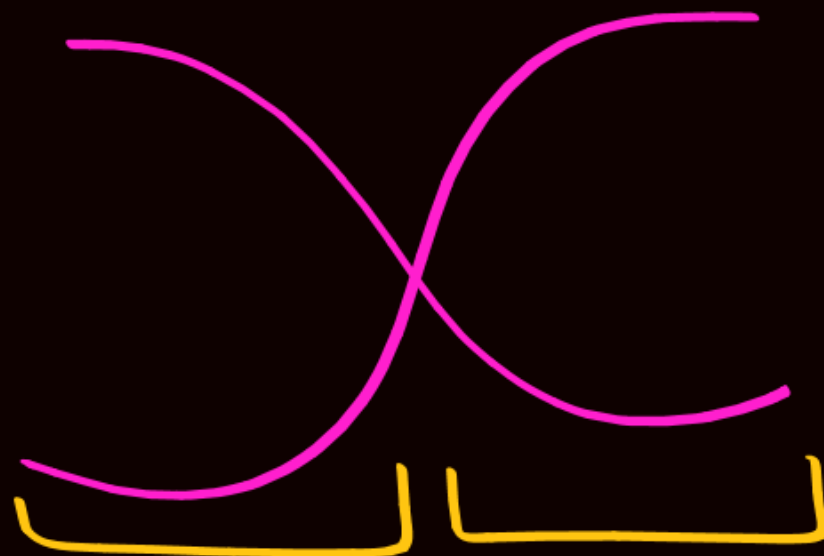
$$\frac{\lambda}{2} + \frac{\cancel{2}\lambda}{\cancel{4}_2}$$

$$\frac{\lambda}{2} + \frac{\lambda}{2}$$

$$\frac{\lambda + \lambda}{2} = \frac{\cancel{2}\lambda}{\cancel{2}} = \textcircled{\lambda} \checkmark$$



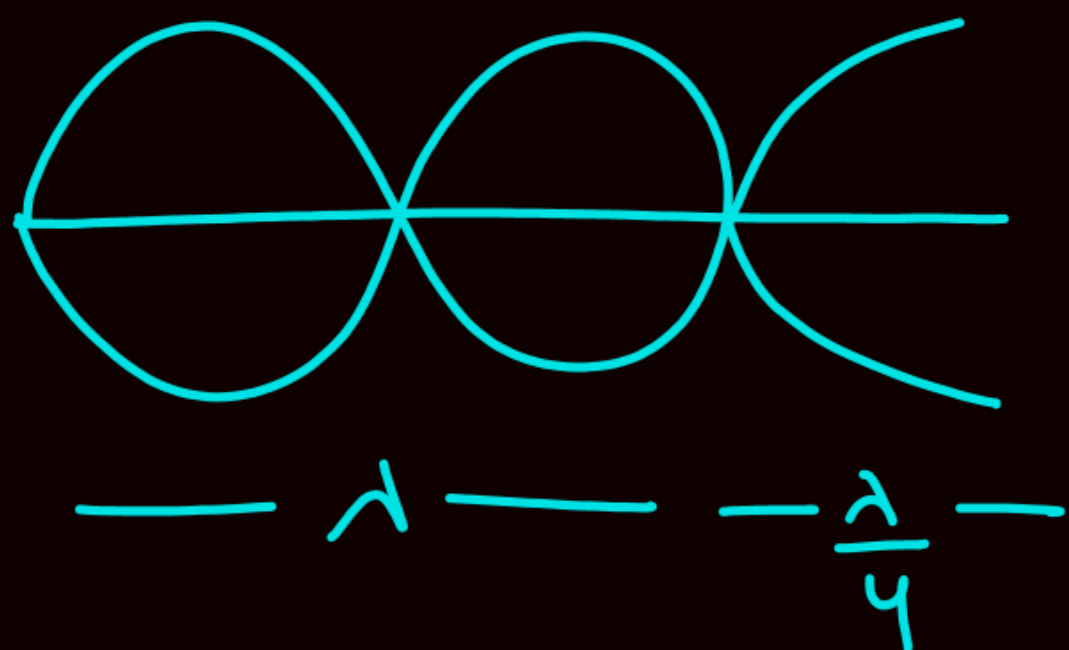
$$l = \frac{\lambda}{2}$$



$$\frac{\lambda}{4} + \frac{\lambda}{4}$$

$$\frac{\lambda}{4} + \frac{\lambda}{4}$$

$$\frac{\lambda + \lambda}{4} = \frac{2\lambda}{4} = \frac{\lambda}{2}$$



$$r + \frac{r}{5}$$

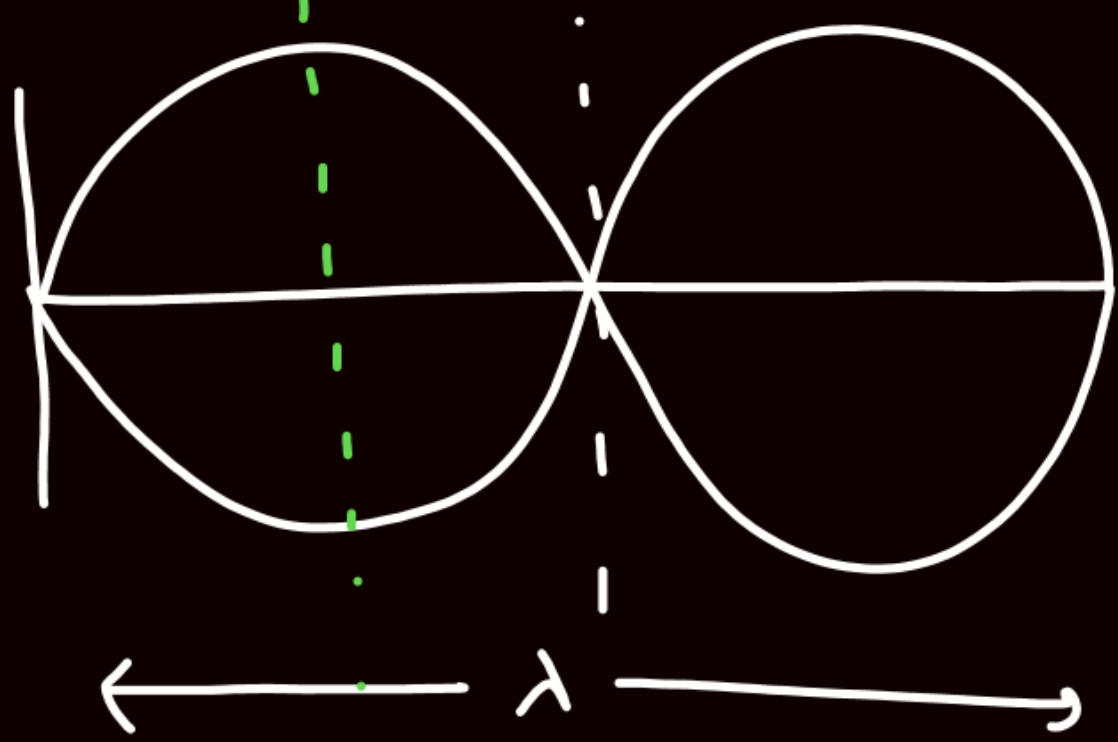
$$\frac{4r + r}{5} = \frac{5r}{5}$$



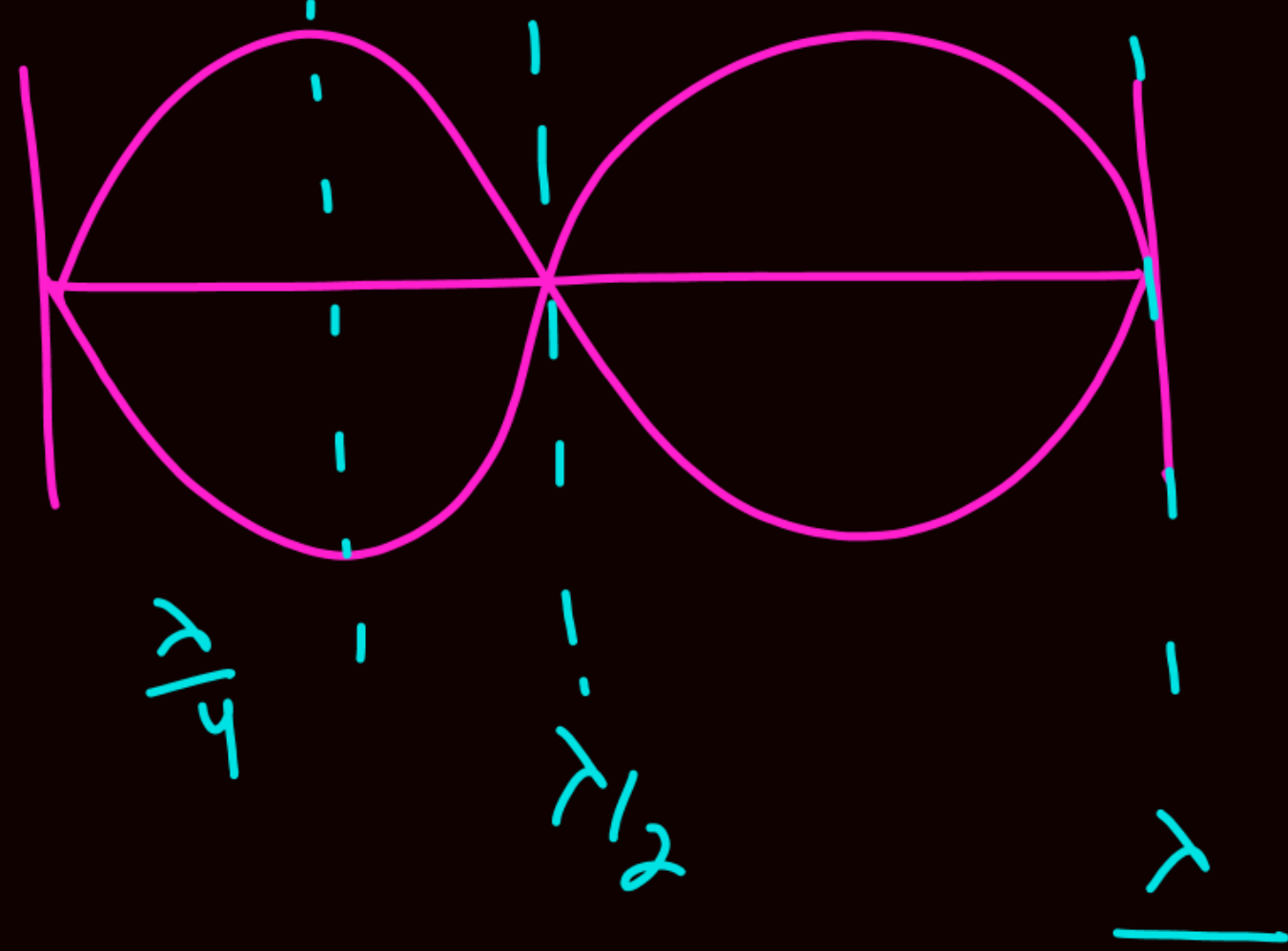
$$\frac{\lambda}{2} + \frac{\lambda}{4}$$

$$\frac{2 \times \lambda}{2 \times 2} + \frac{\lambda}{4}$$

$$\frac{2\lambda}{4} + \frac{\lambda}{4} = \frac{2\lambda + \lambda}{4} = \frac{3\lambda}{4} = l$$



$$\frac{r}{2} \times \frac{r}{2} = \frac{r}{2} \times \frac{r}{2} = \frac{r^2}{4}$$



Mehnat Karo Mehnat karo
Mehnat karne se ho jata hai
Insallah apke 498 No. Aayenge
Bas Yhi motivation apne dil me
rkho bs



Thank You



ala Wallan